

Arrangements of Lines and Algebraic Surfaces

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Introduction

In recent years the Chern numbers c_1^2 and c_2 of algebraic surfaces have aroused special interest. For a minimal surface of general type they are positive and satisfy the inequality $c_1^2 \leq 3c_2$ (see Miyaoka [22] and Yau [31]) where the equality sign holds if and only if the universal cover of the surface is the unit ball $|z_1|^2 + |z_2|^2 < 1$ (see Yau [31] and [23] §2 for the difficult and [11] for the easy direction of this equivalence). It is interesting to know which positive rational numbers ≤ 3 occur as c_1^2/c_2 for a minimal surface of general type. For a long time (before 1955) it was believed that $c_1^2/c_2 \leq 2$, in other words that the signature of a surface of general type is nonpositive. It is interesting to find surfaces with $2 < c_1^2/c_2 \leq 3$. Some were found by Kodaira [19]. Other authors constructed more surfaces of this kind (Holzapfel [13], [14], Inoue [16], Livné [21], Mostow-Siu [25], Miyaoka [23]). Recently Mostow [24] and Deligne-Mostow [5] used a paper of E. Picard of 1885 to construct discrete groups of automorphisms of the unit ball leading to interesting surfaces. In fact, the author was stimulated by Mostow's lecture at the Arbeitstagung 1981 to study the surface Y_1 (see §3.2) which is related to one of the surfaces of Mostow.

For a surface of general type there is a unique minimal model in its birational equivalence class. The Chern numbers of the minimal model are therefore invariants of the birational equivalence class, i.e., they are invariants of the field of meromorphic functions on the surface. In the present paper we study surfaces whose function fields are Kummer extensions $K(a_1^{1/n}, a_2^{1/n}, \dots, a_{k-1}^{1/n})$ of exponent n of the rational function field, i.e., of the function field $K = \mathbb{C}(z_1/z_0, z_2/z_0)$ of the complex projective plane ($a_i \in K$). We consider the following Kummer extensions: For k distinct lines $l_1 = 0, l_2 = 0, \dots, l_k = 0$ in the plane (where l_i is a linear form in z_0, z_1, z_2) we take $a_1 = l_2/l_1, a_2 = l_3/l_1, \dots, a_{k-1} = l_k/l_1$. Under some

assumptions on the arrangement of lines these Kummer extensions ($n \geq 2$) determine surfaces of general type. For three special arrangements of lines we obtain a minimal surface of general type with $c_1^2/c_2 = 3$. Therefore the unit ball appears as an infinite branched cover of the projective plane (with points blown up), the branching locus consisting of lines and the exceptional curves coming from the blown-up points. We also get surfaces with c_1^2/c_2 near to 3, but not arbitrarily close to 3. We show for example that $5/2$ is an accumulation point of accumulation points of the values c_1^2/c_2 for the minimal models of our Kummer extensions.

The topology of the complement of an arrangement of lines in the projective plane is very interesting ([1], [3], [26]), the investigation of the fundamental group of the complement very difficult. We need very little information on the complement (the Euler number is sufficient), because the unramified coverings of the complement occurring in this paper are always abelian. The real simplicial arrangements play an important role in the present paper and it should be mentioned that the complement in the complex projective plane of the complexification of such an arrangement is an Eilenberg-MacLane space [4]. There exists much work on real simplicial arrangements (see Grünbaum [7], [8]), but the classification of these arrangements does not seem to be known.

The inequality $c_1^2 \leq 3c_2$ gives theorems on arrangements of lines (see §3.1) which up to now could not be proved directly.

§1 Arrangements of Lines

We consider the complex projective plane $P_2(\mathbb{C})$ with homogeneous coordinates $z_0 : z_1 : z_2$. An arrangement of k lines is a set of k distinct lines in $P_2(\mathbb{C})$. They can be given by linear forms $l_1, \dots, l_k$ in z_0, z_1, z_2 . The union of the lines is the reducible curve

$$l_1 l_2 \dots l_k = 0.$$

Let t_r ($r \geq 2$) be the number of r -fold points of this curve, i.e., the number of points lying on exactly r lines of the arrangement. Then we have

$$\frac{k(k-1)}{2} = \sum t_r \frac{r(r-1)}{2}$$

For the whole paper *we shall always assume that $k \geq 3$ and $t_k = 0$* . (The arrangement should not be a "pencil", i.e., not all lines should pass through one point.)

1.1 Real arrangements

If all linear forms $l_1, \dots, l_k$ have real coefficients, then $l_1 l_2 \dots l_k = 0$ is an arrangement of lines in the real projective plane $P_2(\mathbb{R})$ which, of course, can also be regarded as an arrangement in $P_2(\mathbb{C})$. A real arrangement defines a cellular decomposition of $P_2(\mathbb{R})$, each cell is bounded by an r -gon. Let p_r be the number of cells bounded by r -gons. Then $p_2 = 0$ (because of our assumption that we do not have a pencil). Let f_0 be the number of vertices, f_1 the number of edges, f_2 the number of cells. The Euler-Poincaré characteristic of $P_2(\mathbb{R})$ equals 1, so

$$(1) \quad f_0 - f_1 + f_2 = 1$$

It is easy to see that

$$(2) \quad f_1 = \sum r \cdot t_r = \frac{1}{2} \sum r \cdot p_r$$

Since $f_0 = \sum t_r$ and $f_2 = \sum p_r$, we conclude from (1) that

$$(3) \quad 3 + \sum (r-3)t_r + \sum (r-3)p_r = 0$$

Since $p_2 = 0$, we have

$$t_2 \geq 3.$$

An arrangement is called simplicial if $p_r = 0$ for $r > 3$. By (3)

$$\text{simplicial} \Leftrightarrow 3 + \sum (r-3)t_r = 0.$$

For a simplicial arrangement

$$(4) \quad f_1 = 3f_0 - 3, \quad f_2 = 2f_0 - 2.$$

If we dualize an arrangement of lines, we get an *arrangement of points* which are not collinear and t_r is the number of lines containing exactly r of the points, t_2 is the number of the so-called ordinary lines. In this setup, Sylvester claimed (1893) that $t_2 > 0$ for real arrangements. The above proof for $t_2 \geq 3$ is due to Melchior (1940). For the history of this problem see Grünbaum's book [8] which is our standard reference for real arrangements. It was proved recently by Sten Hansen [9] that

$$(5) \quad t_2 \geq \left\lfloor \frac{k}{2} \right\rfloor$$

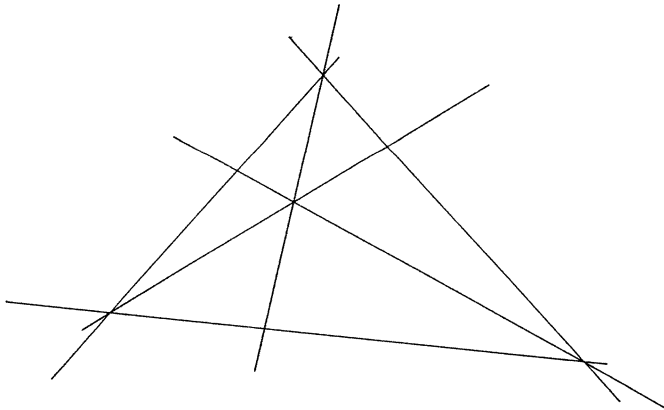
Grünbaum ([7],[8]) has tried to classify *simplicial real arrangements*. There is first the near-pencil $A_0(k)$ which consists of k lines of which $k - 1$ pass through one point ($t_{k-1} = 1$, $t_2 = k - 1$, $t_r = 0$ otherwise). Then he defines the regular arrangements $A_1(2m)$ for $m \geq 3$ and $A_1(4m + 1)$ for $m \geq 2$. We quote from Grünbaum [8]: The regular arrangement $A_1(2m)$ consists of $2m$ lines, of which m are the lines determined by the edges of a regular m -gon in the Euclidean plane, while the other m are the lines of symmetry of that m -gon. The regular arrangement $A_1(4m + 1)$ consists of $4m + 1$ lines of which $4m$ form the arrangement $A_1(4m)$ while the last one is the line at infinity. Grünbaum ([7],[8]) conjectures that up to *combinatorial equivalence* there are besides the three series $A_0(k)$, $A_1(2m)$, $A_1(4m + 1)$ exactly 91 simplicial real arrangements. He lists 90 of them in [7] and gives the extra one in [8]. He lists them (together with the three series) as $A_i(k)$ where k is the number of lines and i a counting index. Grünbaum also gives the values of the t_r . For $A_1(2m)$ we have (for $m > 3$)

$$t_2 = m, t_3 = \frac{m(m-1)}{2}, t_m = 1, t_r = 0 \text{ otherwise.}$$

For $A_1(4m + 1)$ we have (for $m > 2$)

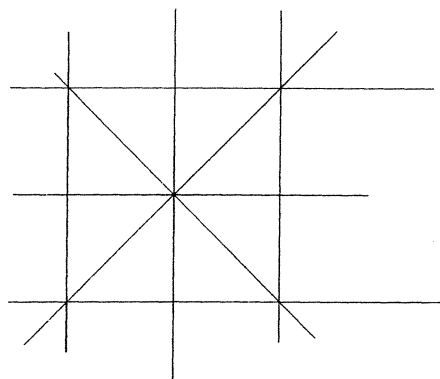
$$t_2 = 3m, t_3 = m(2m - 2), t_4 = m, t_{2m} = 1, t_r = 0 \text{ otherwise.}$$

The arrangement $A_1(6)$ is projectively equivalent to any complete quadrilateral



$$t_2 = 3, t_3 = 4, t_r = 0 \text{ otherwise.}$$

The arrangement $A_1(8)$ looks as follows



$$t_2 = 4, t_3 = 6, t_4 = 1, t_r = 0 \text{ otherwise.}$$

We get $A_1(9)$ if we add the infinite line.

For $A_1(9)$ we have

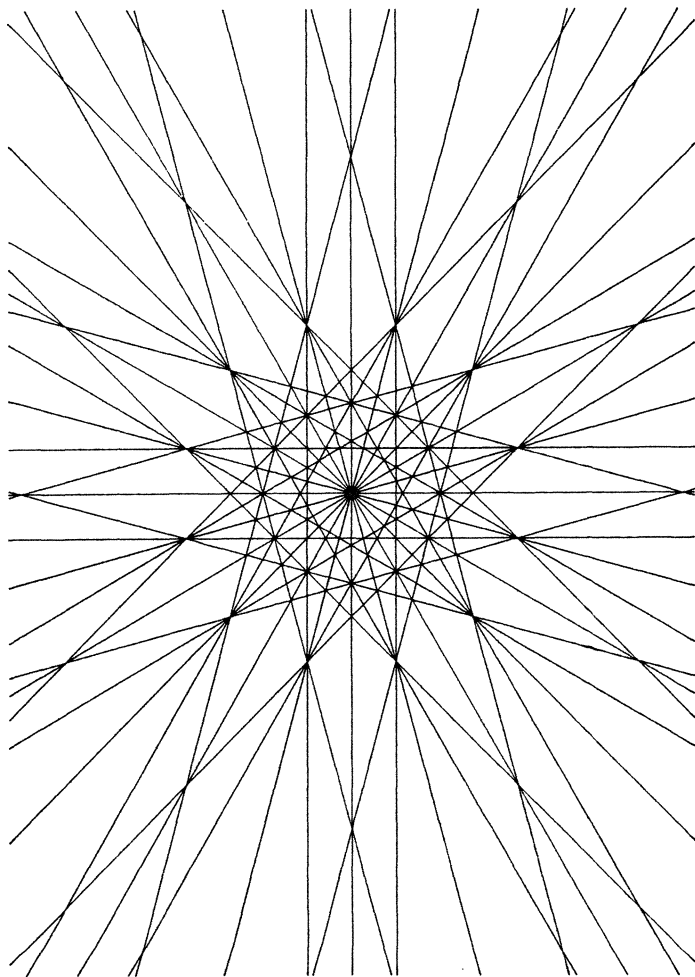
$$t_2 = 6, t_3 = 4, t_4 = 3, t_r = 0 \text{ otherwise.}$$

For $k = 19$ Grünbaum has the following list (the near-pencil is omitted)

	f_0	t_2	t_3	t_4	t_5	t_6
$A_1(19)$	49	21	18	6	0	4
$A_2(19)$	51	21	18	6	6	0
$A_3(19)$	49	24	12	6	6	1
$A_4(19)$	51	20	20	6	4	1
$A_5(19)$	51	20	20	6	4	1
$A_6(19)$	51	20	20	6	4	1
$A_7(19)$	52	21	15	15	0	1

$$t_r = 0 \text{ otherwise}$$

The maximal k for Grünbaum's 91 exceptional arrangements is 37. Besides $A_0(37)$ and $A_1(37)$ belonging to a series he has $A_2(37)$ and $A_3(37)$ where $A_2(37)$ is reproduced here from [7]. (It contains the infinite line.)



$$A_2(37)$$

$t_2 = 72, t_3 = 72, t_4 = 12, t_5 = 24, t_{12} = 1$, all other $t_r = 0$, $f_0 = 181$.

(1.2) Arrangements defined by reflection groups

We first consider real arrangements. A finite reflection group G in the Euclidean space $\mathbb{R}^l$ has finitely many reflecting hyperplanes. Let $L(G)$ be the set of linear subspaces of $\mathbb{R}^l$ which are intersections of reflecting

hyperplanes. If V is a 3-dimensional subspace belonging to $L(G)$, we take the set $A(V)$ of all 2-dimensional elements of $L(G)$ which are contained in V . If we pass from V to the projective plane defined by V , then $A(V)$ defines an arrangement of lines which is simplicial (see Orlik and Solomon [27], [28], and the literature given there). The paper [27] gives a complete list of the arrangements obtainable in this way, however the values of the t_r are not given. But Orlik and Solomon have communicated to me a list of the t_r of these arrangements. J. Tits had mentioned to me before that I should look at such arrangements. He gave me the values of the t_r for one arrangement coming from the Weyl group of E_8 .

Let G be the Weyl group of E_8 acting as reflection group in $\mathbb{R}^8$. By the above method we obtain 8 arrangements of lines. There are two arrangements with 19 lines. They occur as $A_1(19)$ and $A_3(19)$ in Grünbaum's list.

We can take in particular the arrangements defined by reflection groups in $\mathbb{R}^3$, ($V = \mathbb{R}^3$, $k =$ number of reflecting hyperplanes). We have the symmetry groups of the regular tetrahedron, the cube, the icosahedron of orders 24, 48, 120 respectively (improper rotations included). For the tetrahedron and the cube we get the Weyl groups of A_3 and B_3 and the arrangements $A_1(6)$ and $A_1(9)$ mentioned in 1.1. For the icosahedron we get the arrangement $A_1(15)$ of [7] with $t_2 = 15$, $t_3 = 10$, $t_5 = 6$, $t_r = 0$ otherwise.

If we consider finite unitary reflection groups G in the hermitian vector space $\mathbb{C}^l$, we can consider $L(G)$ and $A(V)$ as before in the real case and obtain interesting arrangements of lines (see [28]). Orlik and Solomon have communicated to me a complete list with the values of the t_r .

The groups $G(m, p, n)$ (see the classification in Shephard and Todd [30]) give rise to the following arrangements [28] ($m \geq 2$)

name	k	t_2	t_3	t_m	t_{m+1}	t_{m+2}
$A_3^0(m)$	$3m$	0	m^2	3	0	0
$A_3^1(m)$	$3m + 1$	m	m^2	1	2	0
$A_3^2(m)$	$3m + 2$	$2m$	m^2	0	2	1
$A_3^3(m)$	$3m + 3$	$3m$	m^2	0	0	3

All other $t_r = 0$. We do not say more about these arrangements here, because they occur in some other context in 1.3. For $m = 2, 3$ the above table has to be read in an obvious way, e.g., $A_3^0(3)$, $k = 9$, $t_2 = 0$, $t_3 = 12$, all other $t_r = 0$. For $m = 2$ we get the real arrangements $A_1(6)$, $A_1(7)$, $A_1(8)$, $A_1(9)$ of [7]. (Compare 1.1).

In the following table we list only the arrangements coming from irreducible unitary reflection groups acting on $\mathbb{C}^3$ (omitting the $G(m, p, 3)$ and the real groups). The groups have "registration numbers" according to the Shephard-Todd classification [30].

S-T number	$ G $	k	t_2	t_3	t_4	t_5	projective
24	336	21	0	28	21	0	G_{168}
25	648	12	12	0	9	0	G_{216}
26	1296	21	36	0	9	12	G_{216}
27	2160	45	0	120	45	36	G_{360}

The corresponding groups of projective transformations are the simple group of order 168 (studied by F. Klein), the automorphism group of the Hesse pencil (see 1.3) of order 216 and the Valentiner-Wiman group of order 360 (isomorphic to the alternating group of six letters, see [6], Volume 2, Anhang). The group G_{168} has 21 elements of order 2. The group G_{360} has 45 elements of order 2. The arrangements consist of the pointwise fixed lines of these 21 (or 45) involutions of $P_2(\mathbb{C})$.

(1.3) Arrangements defined by cubic curves

Cubic curves give us arrangements in the dual setup, where we look for arrangements of points and where t_r denotes the number of lines containing exactly r points. Let C be a smooth cubic curve in $P_2(\mathbb{C})$. The group structure is well-defined once we have chosen one of the 9 inflection points as neutral element. The sum $p_1 + p_2 + p_3$ of $p_1, p_2, p_3 \in C$ equals 0 if and only if $\{p_1, p_2, p_3\}$ is the intersection of C with a line. The inflection points are the points with $3p = 0$. Let U be a finite subgroup of C of order k . Consider the arrangement consisting of the points of U . The k tangents to C in points of U are ordinary lines containing two points of the arrangement provided they are not tangents in inflection points. Therefore $t_2 = k - w$, where w is the number of inflection points contained in U . Since $t_r = 0$ for $r > 3$ and $\frac{k(k-1)}{2} = t_2 + 3t_3$, we have

Lemma. *Let U be a subgroup of order k of a smooth cubic. For this arrangement of k points we have*

$$t_2 = k - w, \quad t_3 = \frac{k(k-1)}{6} + \frac{w}{3}, \quad t_r = 0 \quad \text{otherwise,}$$

where w is the number of inflection points contained in U .

Cyclic groups U of order k can be realized over the reals. Then $w = 1$ or $w = 3$ and $t_3 = \lfloor \frac{k(k-3)}{6} \rfloor + 1$. These arrangements are not simplicial, except for $k = 4$ and $k = 6$. Our basic reference for this construction and many historical remarks about the "orchard problem" is the paper [2] by Burr, Grünbaum and Sloane.

If we take as U the group of the 9 inflection points we get (dually) the arrangement $A_3^0(3)$, see 1.2. The 12 lines responsible for $t_3 = 12$ make up a line arrangement to be mentioned in a moment.

The pencil of cubics passing through the nine inflection points of the smooth cubic C is called Hesse pencil. The Hesse pencil does not depend (up to projective equivalence) on the given cubic. The group of projective automorphisms of $P_2(\mathbb{C})$ which carry the Hesse pencil of the cubic C to itself is the group G_{216} mentioned in 1.2. The Hesse pencil of C has 4 singular cubics consisting of three lines each. These 12 lines are an arrangement with $t_2 = 12$, $t_4 = 9$ ($t_r = 0$ otherwise), the nine quadrupel points being the inflection points. It belongs to the group with No. 25 in the Shephard-Todd classification (see 1.2). If we take these 12 lines together with 9 additional lines E_x to be described we get the arrangement belonging to the group carrying S-T number 26. For each inflection point x we have a line E_x which intersects each cubic C_t of the pencil in $\alpha'_t - x$, $\alpha''_t - x$, $\alpha'''_t - x$, where α'_t , α''_t , α'''_t are the points of order 2 of C_t (and where we use the addition on C_t). Through each vertex of a singular cubic of the pencil pass 3 of these 9 lines.

The construction of point arrangements can also be done with singular cubics. Let C be a singular cubic consisting of three lines C_1, C_2, C_3 defining a triangle with distinct vertices $A_1 = C_2 \cap C_3$, $A_2 = C_3 \cap C_1$, $A_3 = C_1 \cap C_2$. Then $C' = C - \{A_1, A_2, A_3\}$ can be given a group structure

$$C' \simeq \mathbb{C}^* \times \mathbb{Z}/3\mathbb{Z}$$

such that three points B_1, B_2, B_3 of C' (with $B_i \in C_i$) are collinear if and only if $B_1 \cdot B_2 \cdot B_3 = 1$ (Theorem of Menelaos). If in such a group structure we choose the $3m$ points of C' which as points of $\mathbb{C}^* \times \mathbb{Z}/3\mathbb{Z}$ have m^{th} -roots of unity as first component, we get the dual version of the arrangement $A_3^0(m)$ of 1.2. If we add one, two, or three of the points A_1, A_2, A_3 to this arrangement we obtain $A_3^1(m)$, $A_3^2(m)$, $A_3^3(m)$. In a similar way the arrangement $A_1(2m)$ of 1.1 can be obtained by choosing a singular cubic consisting of a conic and a line.

§2 Algebraic Surfaces Associated to Arrangements

For an arrangement $l_1 = 0, \dots, l_k = 0$ ($k \geq 3, t_k = 0$) of lines and a natural number $n \geq 2$ we consider the function field

$$\mathbb{C}(z_1/z_0, z_2/z_0)((l_2/l_1)^{1/n}, (l_3/l_1)^{1/n}, \dots, (l_k/l_1)^{1/n})$$

which is an abelian extension of the function field $\mathbb{C}(z_1/z_0, z_2/z_0)$ of $P_2(\mathbb{C})$ of degree n^{k-1} and Galois group $(\mathbb{Z}/n\mathbb{Z})^{k-1}$. (Of course, it does not matter which line of the arrangement is given the subscript 1.) The function field determines an algebraic surface X with normal singularities which ramifies over the plane with the arrangement as locus of ramification:

$$\pi: X \rightarrow P_2^*(\mathbb{C}).$$

If the point $p \in P_2(\mathbb{C})$ lies on r lines of the arrangement ($r \geq 0$), then $\pi^{-1}(p)$ consists of n^{k-1-r} points which are an orbit of the Galois group $(\mathbb{Z}/n\mathbb{Z})^{k-1}$ acting on X . We desingularize all singularities of X in the minimal way (see §2.1) and obtain a smooth algebraic surface Y depending on the arrangement and on n . The surfaces Y will be investigated in this section.

(2.1) Desingularization

If a point of the plane lies on r lines, we choose an affine coordinate system u, v centered at p with a line of the arrangement not passing through p as line at infinity. The local ring of a point of X lying over p is the extension of the regular local ring at p by $(\tilde{l}_1)^{1/n}, \dots, (\tilde{l}_r)^{1/n}$, where $\tilde{l}_1 = 0, \dots, \tilde{l}_r = 0$ are the equations of the lines passing through p written in terms of the u, v coordinate system. We have singular points over p if and only if $r \geq 3$. Such a singularity is resolved by blowing up p in a projective line E_p . The reader should observe the exceptional role of $r = 2$ in all forthcoming calculations.

A singular point q of X over p is blown up into a smooth curve C which covers the projective line E_p and is the "Riemann surface" over E_p with n^{r-1} sheets defined by the functions $(\tilde{l}_2/\tilde{l}_1)^{1/n}, \dots, (\tilde{l}_r/\tilde{l}_1)^{1/n}$ where u, v are now regarded as homogeneous coordinates on E_p . The Euler number $e(C) = 2 - 2g(C)$ is easy to calculate (Hurwitz)

$$(1) \quad e(C) = n^{r-1}(2 - r) + r \cdot n^{r-2}.$$

We apply this process in all singular points q of X and obtain our smooth surface Y . We have natural maps

$$\rho: Y \rightarrow X, \quad \sigma: Y \rightarrow \hat{P}_2, \quad \tau: \hat{P}_2 \rightarrow P_2(\mathbb{C})$$

where $\hat{P}_2$ is the projective plane with all points p with $r \geq 3$ blown up. Of course, $\pi\rho = \tau\sigma$. The map σ is of degree n^{k-1} and ramifies over the proper transforms in $\hat{P}_2$ of the lines of the arrangement and over the curves E_p , each ramification index equals n . Furthermore, σ^*E_p is a divisor in Y consisting of n^{k-1-r} disjoint curves C each with multiplicity n . All these curves are mapped to each other under the Galois group. As far as self-intersection numbers go we have the equation

$$(\sigma^*E_p) \cdot (\sigma^*E_p) = n^{k-1}(E_p \cdot E_p) = -n^{k-1}$$

which implies for each curve C

$$(2) \quad C \cdot C = -n^{r-2}.$$

(2.2) Characteristic numbers

We have defined a smooth algebraic surface Y for each arrangement of lines and any natural number $n \geq 2$. We wish to calculate the Chern numbers c_2, c_1^2 of Y . The number c_2 is the Euler number of Y , whereas c_1^2 is the selfintersection number of a canonical divisor of Y .

For the calculation of c_2 we use the principle that Euler numbers behave like the cardinal numbers of sets. We denote Euler numbers by e . Let $L_1, \dots, L_k$ be the lines of the arrangement, L their union, and $\text{sing } L$ the set of singular points of L . We first calculate the Euler number of the surface X with the set $\text{sing } X$ of its singular points removed. We have

$$\begin{aligned} (3) \quad e(X - \text{sing } X) &= n^{k-1}e(P_2(\mathbb{C}) - L) + n^{k-2}e(L - \text{sing } L) + n^{k-3}t_2 \\ &= n^{k-1}(3 - 2k + \sum (r-1)t_r) + n^{k-2}(2k - \sum rt_r) + n^{k-3}t_2. \end{aligned}$$

Because of (1) we get for the Euler number of Y

$$(4) \quad e(Y) = e(X - \text{sing } X) + \sum_{r \geq 3} n^{k-1-r}t_r(n^{r-1}(2-r) + rn^{r-2})$$

which gives the following

Proposition. *Let Y be the smooth algebraic surface associated to an arrangement of k lines covering the plane with degree n^{k-1} . Then*

$$(5) \quad \begin{aligned} e(Y)/n^{k-3} &= c_2(Y)/n^{k-3} \\ &= n^2(3 - 2k + f_1 - f_0) + 2n(k - f_1 + f_0) + f_1 - t_2. \end{aligned}$$

Here we have used the abbreviations

$$f_1 = \sum r t_r, \quad f_0 = \sum t_r$$

which in the case real arrangements have a geometric meaning (1.1).

We now calculate the other characteristic number $c_1^2(Y)$. For $p \in \text{sing } L$ let r_p be the number of lines passing through p . If $r_p \geq 3$, then we have the exceptional curve E_p in $\hat{P}_2$. For the divisor $L = L_1 + L_2 + \cdots + L_k$ we consider its improper transform (lift) τ^*L to $\hat{P}_2$ (see 2.1.), then

$$\tau^*L - \sum_{\substack{p \in \text{sing } L \\ r_p \geq 3}} r_p E_p$$

is the proper transform of L . Consider as in 2.1. the map $\sigma: Y \rightarrow \hat{P}_2$. The divisors $\sigma^*(\tau^*L - \sum r_p E_p)$ and $\sigma^*(E_p)$ of Y ($p \in \text{sing } L, r_p \geq 3$) are divisible by n . If K is a canonical divisor of $P_2(\mathbb{C})$, then $\tau^*K + \sum E_p$ is a canonical divisor of $\hat{P}_2$. Applying the well-known facts on the behaviour of canonical divisors under ramified coverings we obtain

Lemma. *For a canonical divisor K on $P_2(\mathbb{C})$ consider the divisor (with rational coefficients)*

$$K_Y = \tau^*K + \sum E_p + \frac{n-1}{n} \left(\sum E_p + \tau^*L - \sum r_p E_p \right)$$

on $\hat{P}_2$, the sums being taken over all $p \in \text{sing } L$ with $r_p \geq 3$. Then σ^*K_Y is a canonical divisor of Y .

In $\hat{P}_2$ we have

$$K_Y K_Y = \left(-3 + k \frac{n-1}{n} \right)^2 - \sum_{r \geq 3} t_r \left(1 + \frac{n-1}{n} (1-r) \right)^2$$

which yields for $c_1^2(Y) = n^{k-1}K_Y K_Y$ the following formula

$$(6) \quad c_1^2(Y)/n^{k-3} = ((k-3)n-k)^2 - \sum_{r \geq 2} t_r((r-2)n-r+1)^2 + t_2$$

Using $\frac{k(k-1)}{2} = \sum t_r \frac{r(r-1)}{2}$ and the abbreviations $f_1 = \sum r t_r$ and $f_0 = \sum t_r$ we get

$$(7) \quad c_1^2(Y)/n^{k-3} = n^2(-5k+9+3f_1-4f_0) + 4n(k-f_1+f_0) + f_1-f_0+k+t_2.$$

(2.3) Rough classification I

It is not difficult to determine the canonical dimension of the surfaces Y and to carry out the rough classification in the sense of Enriques-Kodaira [20]. We first remark without proof that for an arrangement of k lines ($k \geq 4$) in general position (i.e., $t_r = 0$ for $r \geq 3$) the surface Y (covering $P_2(\mathbb{C})$ with degree n^{k-1}) is a complete intersection of $k-3$ Fermat surfaces of degree n in $P_{k-1}(\mathbb{C})$ where a Fermat surface is given by an equation $\sum_{i=0}^{k-1} a_i z_i^n$.

The Galois group of Y over $P_2(\mathbb{C})$ is the obvious automorphism group of such a complete intersection. (For $k=3$, the surface Y is a projective plane.) The characteristic numbers of complete intersections are given by well-known formulas [10] which can be checked with (5) and (6). For $k \geq 6$ and $n \geq 2$ such a complete intersection is of general type (i.e., the canonical dimension equals 2), except for $k=6$ and $n=2$ where we have a $K3$ -surface. We shall prove a similar statement for the surfaces Y belonging to arrangements which need not be in general position.

For the following discussion we always assume

$$(8) \quad k \geq 6, \quad n \geq 2, \quad t_k = t_{k-1} = t_{k-2} = 0.$$

The curve E_p in $\hat{P}_2$ is defined only if $r_p \geq 3$. This is always understood in the following formulas. In $\hat{P}_2$ we have to consider the proper transforms L'_j of the lines L_j of the arrangement:

$$L'_j = \tau^* L_j - \sum_{p \in L_j} E_p.$$

We now calculate $K_Y E_p$ and $K_Y L'_j$ (see the lemma in 2.2 and observe that $\tau^* K$ and $\tau^* L_j$ have with all E_p the intersection number 0).

$$(9) \quad \begin{aligned} K_Y E_p &= -1 + \frac{n-1}{n}(r_p - 1) \geq 0 \\ K_Y E_p &= 0 \quad \Leftrightarrow \quad n = 2 \text{ and } r_p = 3. \end{aligned}$$

We have $K_Y L'_j = -3 + \frac{n-1}{n}k - \sum_{p \in L_j} K_Y E_p$.

Formula (9) implies

$$\sum_{p \in L_j} K_Y E_p = \frac{n-1}{n}(k-1) - \frac{n-1}{n} \#\{p \in L_j \mid r_p = 2\} - \#\{p \in L_j \mid r_p \geq 3\}.$$

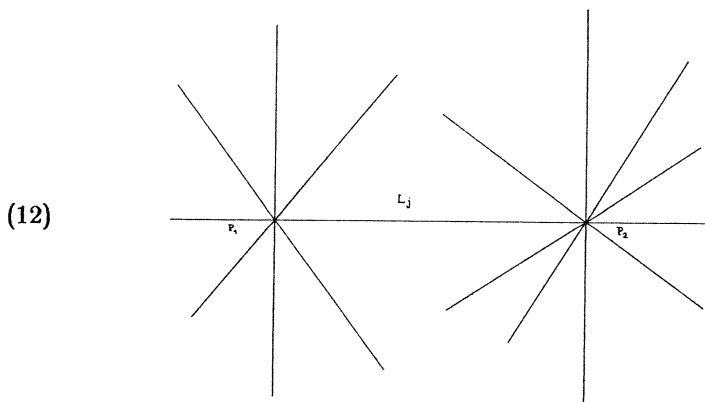
Therefore

$$(10) \quad K_Y L'_j = -2 - \frac{1}{n} + \#\{p \in L_j \mid r_p \geq 2\} - \frac{1}{n} \#\{p \in L_j \mid r_p = 2\}.$$

In view of our assumption (8) it can be checked

$$(11) \quad \begin{aligned} K_Y L'_j &\geq 0 \quad \text{if} \quad \#\{p \in L_j \mid r_p \geq 2\} \geq 3 \\ K_Y L'_j &> 0 \quad \text{if} \quad \#\{p \in L_j \mid r_p \geq 2\} \geq 3 \text{ and } n \geq 3. \end{aligned}$$

If $\#\{p \in L_j \mid r_p \geq 2\} = 2$, then the arrangement looks as follows



$$\begin{aligned} u + v - 1 &= k, \quad u \geq 4, \quad v \geq 4 \\ t_u &= t_v = 1, \quad t_2 = (u-1)(v-1), \quad t_r = 0 \quad \text{otherwise.} \end{aligned}$$

We have $K_Y L'_j = -\frac{1}{n}$.

Theorem. *Assume the arrangement satisfies (8) and is not of type (12). Then the surface is minimal, i.e., does not contain exceptional curves. For an arrangement of type (12) the divisor $\sigma^*L'_j$ on Y consists of n^{k-3} disjoint exceptional curves (each with multiplicity n); blowing them down gives a minimal surface Y_0 which is a product of curves C_1, C_2 with Euler numbers*

$$e(C_1) = n^{u-1}(2-u) + un^{u-2}, \quad e(C_2) = n^{v-1}(2-v) + vn^{v-2}.$$

All the surfaces Y belonging to an arrangement satisfying (8) are of general type for $n \geq 3$. For $k = 6$ and $n = 2$ the surface Y is a $K3$ -surface and for $k \geq 7$ and $n = 2$ it is elliptic (of canonical dimension ≥ 0) or of general type.

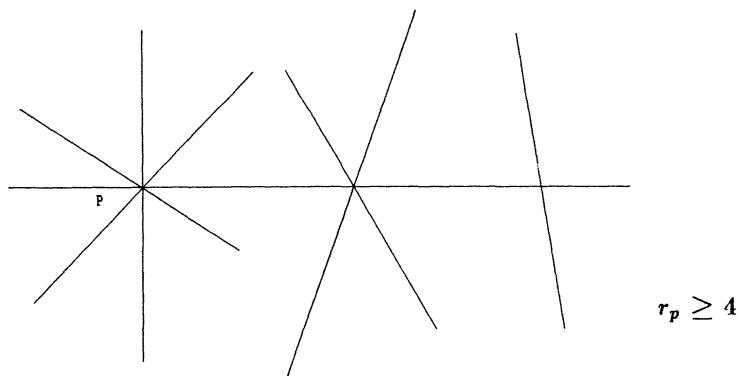
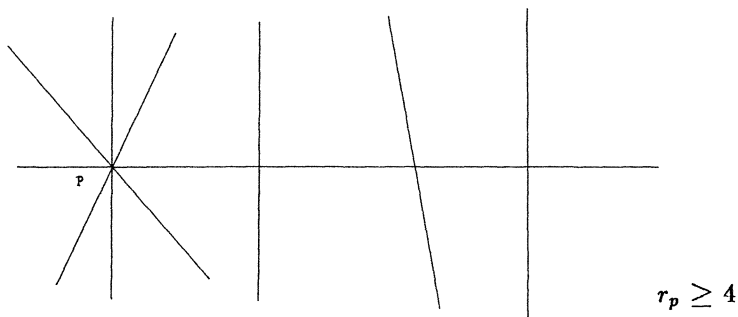
Proof. The divisor K_Y of the lemma in 2.2 can be written as

$$K_Y = \tau^*K + \left(2 - \frac{1}{n}\right) \sum E_p + \frac{n-1}{n} \sum L'_j.$$

We want to choose K such that K_Y is effective. Since $t_k = t_{k-1} = t_{k-2} = 0$, we can find six lines $L_1, L_2, \dots, L_6$ in the arrangement such that not more than 3 of them pass through one point. For K on $P_2(\mathbb{C})$ we take $-\frac{1}{2}(L_1 + L_2 + \dots + L_6)$ allowing half-integral coefficients (multiplying by 2 gives a double canonical divisor). Then K_Y is effective and positive, except for $k = 6$ and $n = 2$ where it vanishes (over the rationals). [In fact, in this case Y is a $K3$ -surface, because the original surface X has only rational double points (of type A_1), their number being $4t_3$, and by Brieskorn's theory Y is a deformation of the surface which we get if we have an arrangement in general position, but then Y is a $K3$ -surface.] The divisor σ^*K_Y is an effective canonical divisor of Y (with rational coefficients). Therefore it contains all exceptional curves of Y . For an exceptional curve F on Y we have $\sigma^*K_Y \cdot F = -1$. Since K_Y is a linear combination of the E_p and L'_j , the exceptional curve F is a component of some divisor σ^*E_p or $\sigma^*L'_j$ (all components of such a divisor being transforms of each other under the Galois group). Therefore $\sigma^*K_Y \cdot F < 0$ is impossible because of (9) and (11) except for arrangements (12). All statements on an arrangement (12) can be checked easily, and we remark that $C_1 \times C_2$ is of general type, except if $n = 2$ and u or v equals 4. In the latter case C_1 or C_2 is elliptic. We still have to prove that Y is of general type if $n \geq 3$ (we can exclude arrangements (12)). We must show that $K_Y K_Y > 0$. But for $n \geq 3$ the

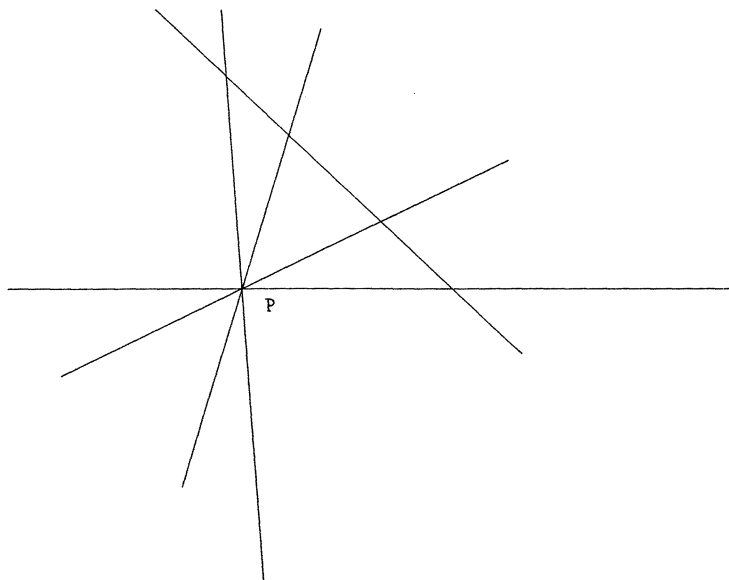
divisor K_Y contains all curves E_p and L'_j with positive multiplicity, and $K_Y K_Y > 0$ follows from (9) and (11). The assertions for $n = 2$ follow because K_Y is effective for $k \geq 7$.

Remark. For $n = 2$ and $k \geq 7$ the surface Y is elliptic if and only if the arrangement is one of the following or of type (12) with u or v equal to 4.



(2.4) Rough classification II (for $t_{k-1} = 1$ or $t_{k-2} = 1$, and $k \geq 6$)

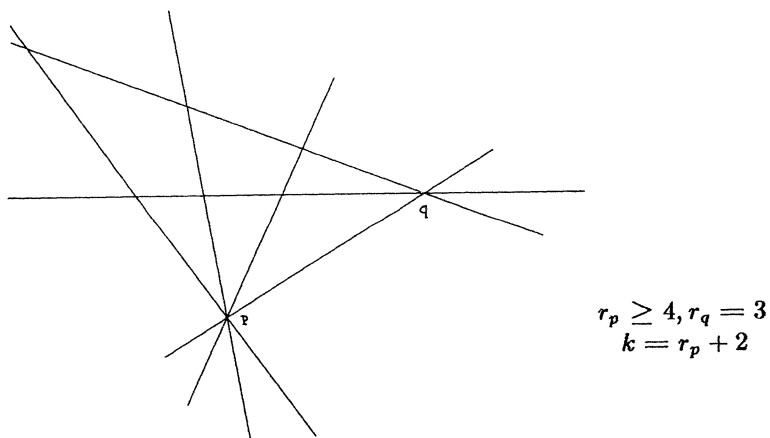
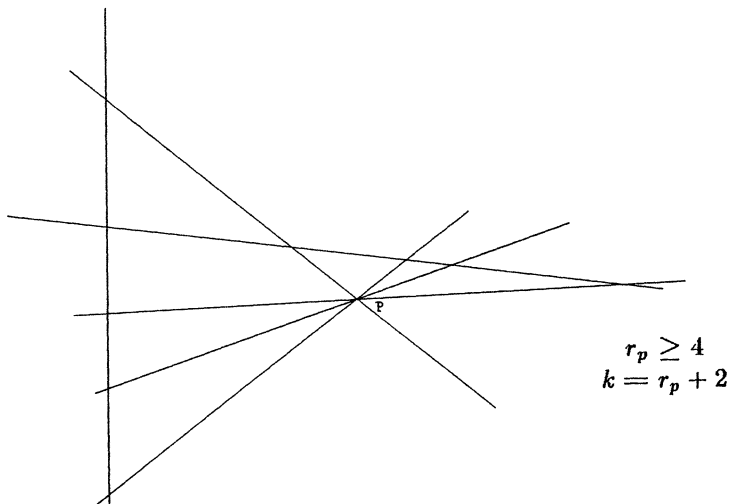
An arrangement



$$\begin{aligned} r_p &\geq 2 \\ k &= r_p + 1 \end{aligned}$$

is called a near-pencil [8]. Then the surface Y is birationally equivalent to a ruled surface for all $n \geq 2$ and has canonical dimension $-\infty$.

The arrangements



lead to surfaces Y which for $n = 2$ are again birationally equivalent to ruled surfaces and have canonical dimensions $-\infty$. For $n = 3$ we get elliptic

surfaces with canonical dimension ≥ 0 and for $n \geq 4$ surfaces of general type.

The theorem in 2.3 and the above remarks complete the rough classification of the surfaces Y for all arrangements with $k \geq 6$. In particular, we have

Theorem. *For an arrangement with $k \geq 6$ and $t_k = t_{k-1} = 0$, the surface Y has canonical dimension ≥ 0 for $n \geq 3$. For an arrangement with $k \geq 6$ and $t_k = t_{k-1} = t_{k-2} = 0$ the canonical dimension is ≥ 0 for $n \geq 2$. (In these cases no ruled surfaces occur.)*

We leave it to the reader to investigate the surfaces for $k = 4, 5$ ($t_k = t_{k-1} = 0$). The surface Y is rational, or of canonical dimension ≥ 0 .

(2.5) The Miyaoka-Yau inequality

We consider the Chern numbers c_1^2 and c_2 for an algebraic surface S . If S is of general type and minimal, then

$$(13) \quad c_1^2(S) \leq 3c_2(S),$$

and equality holds if and only if the universal cover of S is biholomorphically equivalent to the unit ball $|z_1|^2 + |z_2|^2 < 1$ (compare the introduction). If (13) holds for a surface S , then it is also true for any surface obtained by blowing up points of S . Therefore (13) is true for all surfaces of general type. The Enriques-Kodaira classification gives also a list of minimal algebraic surfaces. The inequality (13) is wrong for the minimal algebraic surface S if and only if S is a ruled surface with a curve of genus $g \geq 2$ as base curve. In this case $c_1^2 = 4(1 - g)$, $c_2 = 8(1 - g)$.

Therefore, (13) is true if S is not birationally equivalent to a ruled surface with base curve of genus $g \geq 2$. In particular, (13) is true if the canonical dimension of S is ≥ 0 or if the surface is rational.

The results of 2.3 and 2.4 imply

Theorem. *For an arrangement of k lines with $t_k = t_{k-1} = t_{k-2} = 0$ the algebraic surface Y satisfies the Miyaoka-Yau inequality for all $n \geq 2$. If $t_k = t_{k-1} = 0$ the inequality is true for all $n \geq 3$.*

§3 The Miyaoka-Yau Inequality and Applications

For an algebraic surface S of general type the Euler number $c_2(S)$ is positive, the ratio $c_1^2(S)/c_2(S)$ is less or equal 3. The signature is positive if and only if $c_1^2(S)/c_2(S)$ is greater than 2. We want to construct surfaces using arrangements of lines for which the ratio is 3 or near to 3. In any case the whole range between 2 and 3 is interesting. For the surfaces coming from arrangements of lines the number $3c_2(S) - c_1^2(S)$ is calculated in terms of the combinatorial data of the arrangement. Since $3c_2(S) - c_1^2(S) \geq 0$, this gives restrictions on the combinatorial data and results on arrangements which could not be proved directly up to now.

(3.1) The quadratic polynomial associated to an arrangement

Let Y be the smooth algebraic surface associated to an arrangement of k lines covering the plane with degree n^{k-1} . Then

$$(1) \quad (3c_2(Y) - c_1^2(Y))/n^{k-3} = n^2(f_0 - k) + n(-2f_1 + 2f_0 + 2k) + 2f_1 + f_0 - k - 4t_2.$$

This follows from equations (5) and (7) in §2. If we replace in (1) the “variable” n by $x + 1$ we obtain the following quadratic polynomial

$$(2) \quad F(x) = x^2(f_0 - k) - 2x(f_1 - 2f_0) + 4(f_0 - t_2).$$

The polynomial F is called the *quadratic polynomial of the arrangement*. The coefficients $f_1 - 2f_0$ and $f_0 - t_2$ are positive except if the arrangement is in general position when both vanish. The coefficient $f_0 - k$ is positive if $t_k = t_{k-1} = 0$. This follows from the theorem in 2.5. The inequality $f_0 \geq k$ (if $t_k = 0$) is an old result on arrangements of lines valid in every projective plane [8].

The theorem in 2.5 implies

Theorem. *For an arrangement of k lines in the complex projective plane we consider the associated quadratic polynomial $F(x)$. If $t_k = t_{k-1} = t_{k-2} = 0$, then $F(x) \geq 0$ for all integers x . If $t_k = t_{k-1} = 0$, then $F(x) \geq 0$ for all integers $x \neq 1$; in particular $F(2) \geq 0$ which means*

$$(3) \quad t_2 + t_3 \geq k + t_5 + 2t_6 + 3t_7 + \cdots.$$

The inequality (3) shows for example that arrangements of k lines with $t_2 = t_3 = t_k = 0$ are impossible in the complex projective plane.

Conjecture. *The quadratic form*

$$(f_0 - k)u^2 - 2(f_1 - 2f_0)uv + 4(f_0 - t_2)v^2$$

is positive semidefinite if $t_k = t_{k-1} = t_{k-2} = 0$, in other words we conjecture

$$(4) \quad (f_1 - 2f_0)^2 \leq 4(f_0 - k)(f_0 - t_2).$$

The conjecture can be verified for all arrangements mentioned in §1. For a real simplicial arrangement we have $f_1 = 3f_0 - 3$ and

$$(5) \quad F(x) = x^2(f_0 - k) - 2x(f_0 - 3) + 4(f_0 - t_2).$$

The conjecture claims (if $t_k = t_{k-1} = t_{k-2} = 0$)

$$(6) \quad (f_0 - 3)^2 \leq 4(f_0 - k)(f_0 - t_2).$$

Of course, we are interested in cases when $F(x) = 0$ for some natural number x . Then we get a surface with $c_1^2 = 3c_2$. Up to now only the following arrangements could be found

I. The real simplicial arrangement $A_1(6)$ (see §1) with $k = 6$, $f_0 = 7$, $t_2 = 3$ and

$$F(x) = (x - 4)^2.$$

II. The arrangement of the 12 lines in the Hesse pencil (see 1.2, S-T number 25, and 1.3) with $k = 12$, $t_2 = 12$, $t_4 = 9$ and

$$F(x) = 9(x - 2)^2.$$

III. The arrangement $A_3^9(3)$ coming from the nine inflection points of a cubic by dualizing. In this case $k = 9$, $t_3 = 12$ and

$$F(x) = 3(x - 4)^2$$

Arrangement II is the only arrangement known with equality sign in (3).

(3.2) Three surfaces with $c_1^2 = 3c_2$

Using the arrangements I, II, III in 3.1 we get three surfaces Y_1, Y_2, Y_3 with $c_1^2 = 3c_2$, they are coverings of the plane of degrees 5^5 , 3^{11} , 5^8 respectively. (Recall that $n = x + 1$.) The Euler numbers c_2 can be calculated by §2(5).

$$c_2(Y_1) = 15 \cdot 5^3, \quad c_2(Y_2) = 48 \cdot 3^9, \quad c_2(Y_3) = 111 \cdot 5^6.$$

Ishida [17] has calculated the irregularity of these surfaces. He obtained

$$q(Y_1) = 30, \quad q(Y_2) = 154, \quad q(Y_3) = 96.$$

Inoue [16] and Livné [21] constructed several surfaces of general type with $c_1^2 = 3c_2$ using elliptic modular surfaces. For convenience we copy the explanation of these surfaces from Ishida [18]: Let $E(n)$ be the elliptic modular surface of level $n > 3$. The elliptic fibration $E(n) \rightarrow X(n)$ has n^2 sections. The union D of all the sections is divisible in $\text{Pic}(E(n))$ by n if n is odd and by $n/2$ if n is even. Hence for each divisor d of n or $n/2$, we can take a cyclic cover $S = S_d(n)_\Delta \rightarrow E(n)$ of order d which is ramified over D . It is not unique in general, but depends on the choice of the element $\Delta \in \text{Pic}(E(n))$ with $d\Delta = D$. Inoue and Livné showed $c_1(S)^2 = 3c_2(S)$ for the cases

$$(n, d) = (7, 7), (8, 4), (9, 3) \text{ and } (12, 2).$$

In the case $(n, d) = (5, 5)$, Δ is unique and the inverse images of the twenty-five sections of $E(n)$ in S are exceptional curves of the first kind. The nonsingular surface $S(5)$ obtained by contracting all these curves has Chern numbers $c_1^2 = 3c_2 = 225$.

The surface Y_1 admits an action of $A = (\mathbb{Z}/5\mathbb{Z})^5$. There are many subgroups A_2 of A of order 25 which operate freely on Y_1 . Dividing Y_1 by such a subgroup gives a surface with $c_1^2 = 3c_2 = 225$, and A_2 can be chosen in such a way that $Y_1/A_2 = S(5)$ (Hirzebruch and Ishida [18], compare also [12]). Similarly Y_2 admits an action of $A = (\mathbb{Z}/3\mathbb{Z})^{11}$. Then A has a subgroup A_5 of order 3^5 which operates freely, the quotient being $S_3(9)_\Delta$ for the unique $\Delta \in \text{Pic}(E(9))$ such that all automorphisms of $E(9)$ lift to $S_3(9)_\Delta$. This is a result of Ishida [18]. The surface $Y_2/A_5 = S_3(9)_\Delta$ has Chern numbers $c_1^2 = 3c_2 = 16 \cdot 3^6$.

(3.3) Behaviour of c_1^2/c_2 for $n \rightarrow \infty$

If S is minimal surface of general type, then both Chern numbers $c_1^2(S)$ and $c_2(S)$ are positive and we have

$$\frac{1}{5}(c_2(S) - 36) \leq c_1^2(S) \leq 3c_2(S),$$

the first inequality being due to M. Noether. Suppose we have a sequence S_n of minimal surfaces of general type with $\lim_{n \rightarrow \infty} c_2(S_n) = \infty$, then

$$(7) \quad \frac{1}{5} \leq \varliminf \frac{c_1^2(S_n)}{c_2(S_n)} \leq \varlimsup \frac{c_1^2(S_n)}{c_2(S_n)} \leq 3.$$

Problem. Which numbers r in the interval $[\frac{1}{5}, 3]$ occur as

$$r = \lim_{n \rightarrow \infty} \frac{c_1^2(S_n)}{c_2(S_n)}$$

for a sequence S_n of minimal surfaces of general type with $c_1^2(S_n)/c_2(S_n) \neq r$? Let $\mathfrak{R}$ be the set of these r , in other words $\mathfrak{R}$ is the set of accumulation points of the set of values c_1^2/c_2 of minimal surfaces of general type. Which numbers of the interval $[\frac{1}{5}, 3]$ are accumulation points of R ?

For an arrangement of k lines with $k \geq 6$ and $t_k = t_{k-1} = t_{k-2} = 0$ which is not of type (12) (see §2), the surface Y_n associated to the arrangement (covering the plane with degree n^{k-1}) is minimal of general type for $n \geq 3$. Since $c_2(Y_n) > 0$, we have $f_1 - f_0 \geq 2k - 3$ (see the polynomial (5) in §2). Therefore, the linear coefficient of this polynomial is negative and hence $f_1 - f_0 > 2k - 3$. It also follows that the leading coefficient of the polynomial (7) in §2 is positive. We obtain

$$(8) \quad \lim_{n \rightarrow \infty} \frac{c_1^2(Y_n)}{c_2(Y_n)} = \frac{5}{2} - \frac{3f_0 - f_1 - 3}{2(3 - 2k + f_1 - f_0)}$$

where $3 - 2k + f_1 - f_0 > 0$. The limit in (8) will be called the characteristic number γ of the arrangement.

Proposition. The characteristic number γ of a real arrangement is less or equal $5/2$. We have $\gamma = 5/2$ if and only if the real arrangement is simplicial.

Proof. By §1(3) we have $3f_0 - f_1 - 3 = \sum_{r \geq 3} (r - 3)p_r \geq 0$.

Remark. In his paper [15] Iitaka has studied the characteristic number γ from some other point of view. He erroneously claimed $\gamma \leq 5/2$. But this can be wrong for arrangements not definable over the reals, for example $\gamma > \frac{5}{2}$ if $t_2 = 0$.

If the arrangement is in general position ($f_1 = 2f_0$), then $\gamma < 2$, but $\lim_{k \rightarrow \infty} \gamma = 2$. Therefore, the number 2 is an accumulation point (from below) of the set $\mathfrak{R}$. According to [19], $2 + \frac{1}{2n-1} \in \mathfrak{R}$, so 2 is also an accumulation point from above.

For the arrangements mentioned in the lemma of 1.3 we have

$$\gamma = \frac{5}{2} - (k-3-w)/2(k(k-6)/3 - w/3 + 3).$$

Therefore $5/2$ is an accumulation point (from below) of the set $\mathfrak{R}$.

For the arrangement $A_3^0(m)$ of 1.2 we get

$$\gamma = \frac{5}{2} + \frac{3m-6}{2(2m^2-3m)}.$$

Therefore $5/2$ is an accumulation point (from above) of the set $\mathfrak{R}$.

According to Holzapfel[13] the number 3 belongs to $\mathfrak{R}$. We could not prove this using the surfaces coming from line arrangements. However, it is possible to construct surfaces (for some arrangements and some n) with c_1^2/c_2 fairly near to 3.

(3.4) Surfaces with c_1^2/c_2 near to 3

According to §2(5), (7) c_1^2/c_2 is a quotient of quadratic polynomials in n . We give some examples. Let us first take the arrangements I, II, III of 3.1. We obtain

$$\text{I. } 3 - c_1^2/c_2 = (n-5)^2/(2n^2 - 10n + 15) \\ \text{For } n = 6 \text{ we have } c_1^2/c_2 = 2\frac{26}{27}$$

$$\text{II. } 3 - c_1^2/c_2 = 3(n-3)^2/(6n^2 - 18n + 16) \\ \text{For } n = 4 \text{ we have } c_1^2/c_2 = 2\frac{37}{40}$$

$$\text{III. } 3 - c_1^2/c_2 = 3(n-5)^2/(3n^2 - 10n + 12) \\ \text{For } n = 6 \text{ we have } c_1^2/c_2 = 2\frac{59}{60}$$

For the icosahedral arrangement (see 1.2, called $A_1(15)$ in Grünbaum's list) we obtain

$$3 - c_1^2/c_2 = \frac{16n^2 - 88n + 136}{32n^2 - 88n + 75}$$

For $n = 3$ this gives $c_1^2/c_2 = 2\frac{83}{99}$.

For the arrangement $A_1(19)$ with 19 lines arising from the Lie group E_8 (see 1.2) we have

$$3 - c_1^2/c_2 = \frac{30n^2 - 152n + 234}{60n^2 - 152n + 123}$$

For $n = 3$ we have $c_1^2/c_2 = 2\frac{53}{69}$.

For the arrangement with S-T number 24 belonging to the simple group of order 168 we have

$$3 - c_1^2/c_2 = \frac{7(n^2 - 7n + 13)}{20n^2 - 49n + 42}$$

For $n = 4$ we have $c_1^2/c_2 = 2\frac{159}{166}$.

The reader may try some more of the arrangements considered in §1. The resulting surfaces should be investigated more thoroughly, e.g., the irregularities should be calculated. The results of this paper can be generalized in many ways, e.g., one can consider conics instead of lines.

(3.5) Remark

It might be that the set $\mathfrak{A}$ introduced in 3.3 is the whole interval $[\frac{1}{5}, 3]$. In fact, it follows from Persson's results [29] that $[\frac{1}{5}, 2] \subset \mathfrak{A}$. But not much seems to be known for the interval $[2, 3]$. It also occurs in [21] that $5/2$ is an accumulation point of $\mathfrak{A}$.

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Remarks added in proof:

1) The conjecture in 3.1 is wrong as Dr. Werner Meyer pointed out:

Take arrangements A_1 and A_2 of k_1 and k_2 lines respectively which are in general position to each other. Consider the combined arrangement $A_1 \cup A_2$. Then the numerical characteristics t_r are additive except for t_2 which satisfies

$$t_2(A_1 \cup A_2) = t_2(A_1) + t_2(A_2) + k_1 k_2$$

Let F_1, F_2, F be the quadratic polynomials for $A_1, A_2, A_1 \cup A_2$. Then

$$F(x) = F_1(x) + F_2(x) + k_1 k_2 x^2$$

If we take for A_2 a pencil with k_2 large with respect to k_1 , then $F(x)$ becomes indefinite. Maybe the conjecture remains true if one assumes that the arrangement "does not contain large pencils".

2) The inequality (3) in 3.1 can be improved by using results of F. Sakai (Semi-Stable Curves on Algebraic Surfaces and Logarithmic Pluricanonical Maps, Math. Ann 254, p. 89–120 (1980)). We have

$$t_2 + \frac{3}{4}t_3 \geq k + t_5 + 2t_6 + 3t_7 + \dots$$

This is sharp for $A_1(6), A_1(9), A_3^0(3), A_3^0(4)$ and the arrangements with S–T numbers 24 and 25.

3) B. Grünbaum has written to me that the arrangements $A_2(17)$ and $A_7(17)$ of his list are isomorphic.

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